

PM2.5 Concentration Prediction

West Bengal Air Quality Analysis

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Post Graduate Diploma in Business Analytics
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Dataset: CPCB - Central Pollution Control Board, West Bengal
Target Variable: PM2.5 concentration ($\mu\text{g}/\text{m}^3$)

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1. Executive Summary

This report presents a comprehensive regression analysis for predicting PM2.5 concentrations across West Bengal monitoring stations using hourly pollutant and meteorological data sourced from the Central Pollution Control Board (CPCB). Five models were built and evaluated: Ordinary Least Squares Regression on raw features (OLS-Raw), OLS on log-transformed features (OLS-Log), Polynomial Regression of degree 2 with Ridge regularisation, LightGBM, and XGBoost.

LightGBM emerged as the best-performing model with an R^2 of 0.696 and RMSE of 28.40 $\mu\text{g}/\text{m}^3$ on the original PM2.5 scale. XGBoost closely followed at $R^2 = 0.672$. Polynomial Regression ($R^2 = 0.470$) outperformed both OLS variants, confirming that the relationship between pollutants and PM2.5 is nonlinear. SHAP analysis identified NO_2 and CO as the dominant predictors of PM2.5 elevation, while high humidity and wind speed were associated with lower concentrations through pollutant dispersal.

A key preprocessing decision was the removal of PM10 from the feature set. PM10 exhibited a Pearson correlation of 0.92 with PM2.5, which would have constituted near-data leakage and artificially inflated model performance. All remaining features passed multicollinearity checks with VIF values well below 5.

2. Introduction & Problem Statement

Particulate matter with diameter of 2.5 micrometres or less (PM2.5) is one of the most harmful air pollutants due to its ability to penetrate deep into the lungs and bloodstream. West Bengal — and Kolkata in particular — consistently ranks among the most polluted regions in India, with PM2.5 regularly exceeding the WHO guideline of 15 $\mu\text{g}/\text{m}^3$ and India's national standard of 60 $\mu\text{g}/\text{m}^3$ for a 24-hour average.

Accurate prediction of PM2.5 from co-pollutants and meteorological variables enables several practical applications: early warning systems for health advisories, identifying major emission sources, and guiding policy interventions. This study uses monitoring station data from West Bengal to build predictive models ranging from interpretable linear regression to high-performance gradient boosting ensembles.

2.1 Objectives

- Build and compare five regression models to predict PM2.5
- Apply feature engineering including log transforms to address skewness
- Diagnose multicollinearity and correlation structure before modelling
- Evaluate all models consistently on RMSE, MAE, R^2 , and Adjusted R^2 on the original PM2.5 scale
- Interpret the best model using SHAP (SHapley Additive exPlanations) values

3. Dataset Description

The dataset was obtained from CPCB monitoring stations distributed across West Bengal. Each station records hourly averages of air pollutants and meteorological parameters. Multiple CSV files — one per monitoring station — were combined into a single dataframe using station metadata.

3.1 Features

After initial cleaning, the following features were retained for analysis:

Feature	Type	Description
pm25	Target	PM2.5 concentration in $\mu\text{g}/\text{m}^3$ — fine particulate matter ($\leq 2.5 \mu\text{m}$)
no2	Pollutant	Nitrogen Dioxide in $\mu\text{g}/\text{m}^3$
nox	Pollutant	Nitrogen Oxides in ppb
nh3	Pollutant	Ammonia in $\mu\text{g}/\text{m}^3$
so2	Pollutant	Sulphur Dioxide in $\mu\text{g}/\text{m}^3$
co	Pollutant	Carbon Monoxide in mg/m^3
ozone	Pollutant	Ground-level Ozone in $\mu\text{g}/\text{m}^3$
wind_speed	Meteorological	Wind speed in m/s
wind_direction	Meteorological	Wind direction in degrees (merged from two source columns)
humidity	Meteorological	Relative Humidity in %
solar_radiation	Meteorological	Solar Radiation in W/m^2
city_encoded	Location	Label-encoded city identifier

Table 1: Feature set after preprocessing

4. Data Preprocessing

4.1 Missing Value Treatment

Rows where the target variable PM2.5 was null were dropped entirely, as they cannot contribute to supervised learning. For all remaining features, missing values were imputed using the per-city median — this preserves city-level pollution patterns rather than applying a single global fill. Any columns that remained null after city-level imputation (due to a city having no observations for a feature) were filled with the global median as a fallback.

4.2 Outlier Removal

PM2.5 readings below 0 $\mu\text{g}/\text{m}^3$ (physically impossible) and above 500 $\mu\text{g}/\text{m}^3$ (extreme sensor errors) were removed. This conservative threshold retains genuine pollution episodes such as severe winter smog events while eliminating clearly erroneous readings.

4.3 City Encoding

The categorical city column was label-encoded to convert it into a numeric feature. This provides the model with location context to capture systematic city-level differences in baseline pollution levels.

4.4 Removal of PM10

PM10 was initially included in the feature set. However, the Pearson correlation between PM10 and PM2.5 was found to be 0.92 in both the raw and log-transformed feature spaces. Retaining PM10 would introduce a form of data leakage — PM10 is essentially a superset of PM2.5, so any model with PM10 as a feature would be partially predicting the target from a near-identical quantity rather than from independent causal drivers. PM10 was therefore removed from all models.

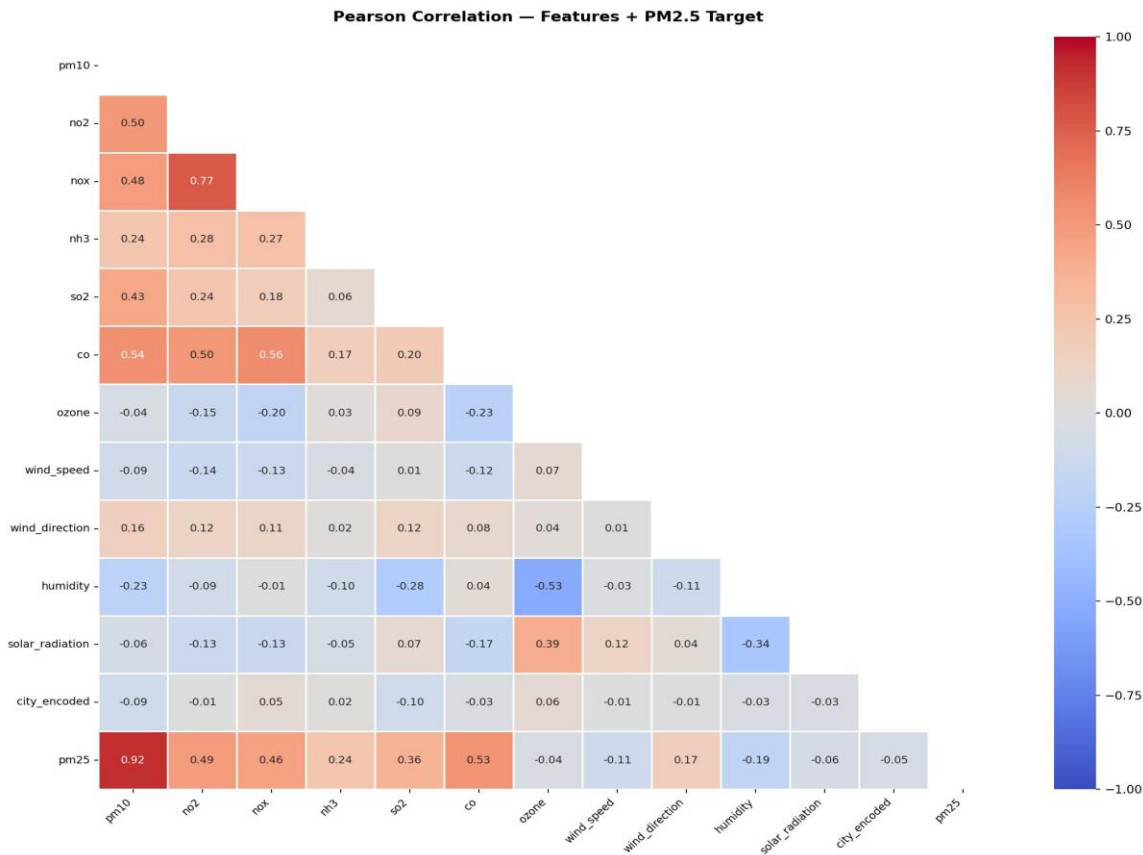


Figure 1: Pearson Correlation Heatmap — Raw features including PM10. Note $r=0.92$ between PM10 and PM2.5.

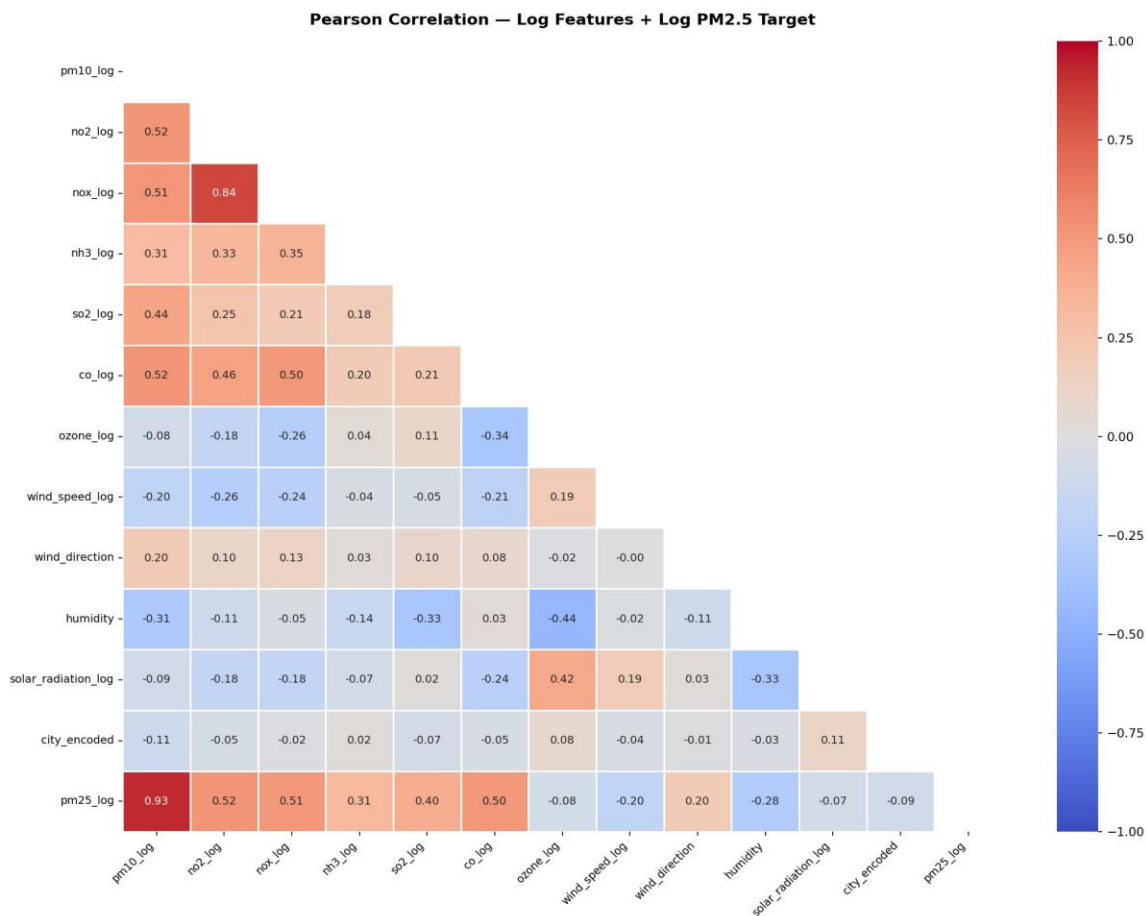


Figure 2: Pearson Correlation Heatmap — Log-transformed features including PM10. $r=0.93$ between $\log(\text{PM10})$ and $\log(\text{PM2.5})$.

5. Feature Engineering

5.1 Log Transformation of Skewed Features

Air pollution measurements are typically right-skewed — most hourly readings are low, but occasionally spike to very high values during pollution events. Applying a $\log(1+x)$ transform (i.e., $\log(1+x)$) compresses the upper tail, making the distribution more symmetric and improving the assumptions of linear regression models.

The following features were log-transformed: NO_2 , NO_x , NH_3 , SO_2 , CO , Ozone, Wind Speed, and Solar Radiation. The target variable $\text{PM}_{2.5}$ was also log-transformed for the OLS-Log model, with predictions back-transformed to the original scale for metric reporting.

5.2 Train / Test Split

All models used a consistent 80/20 train-test split with `random_state=42`. Feature scaling (StandardScaler, fitted on the training set only) was applied before OLS and Polynomial Regression models to ensure comparability of coefficients and to satisfy the assumptions of regularised regression.

6. Pre-Model Diagnostics

6.1 Correlation Analysis — Final Feature Set

After removing PM10, the correlation heatmaps were re-examined. The strongest predictors of PM2.5 are CO ($r = 0.53$), NO₂ ($r = 0.49$), and NO_x ($r = 0.46$), which are primary combustion products. Humidity shows a moderate negative correlation ($r = -0.19$), consistent with wet scavenging of particulates. Log-transforming the features slightly strengthens most linear correlations.

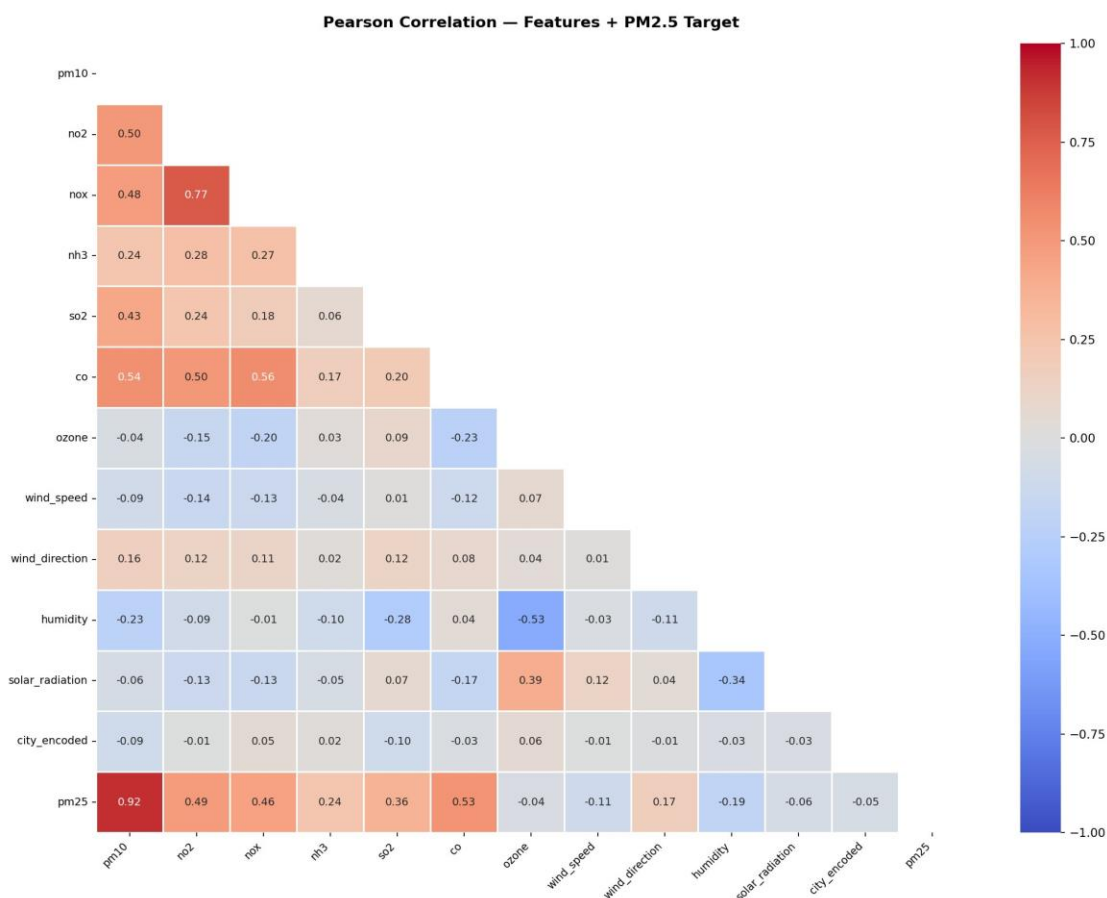


Figure 3: Correlation Heatmap — Raw features (without PM10) vs PM2.5.

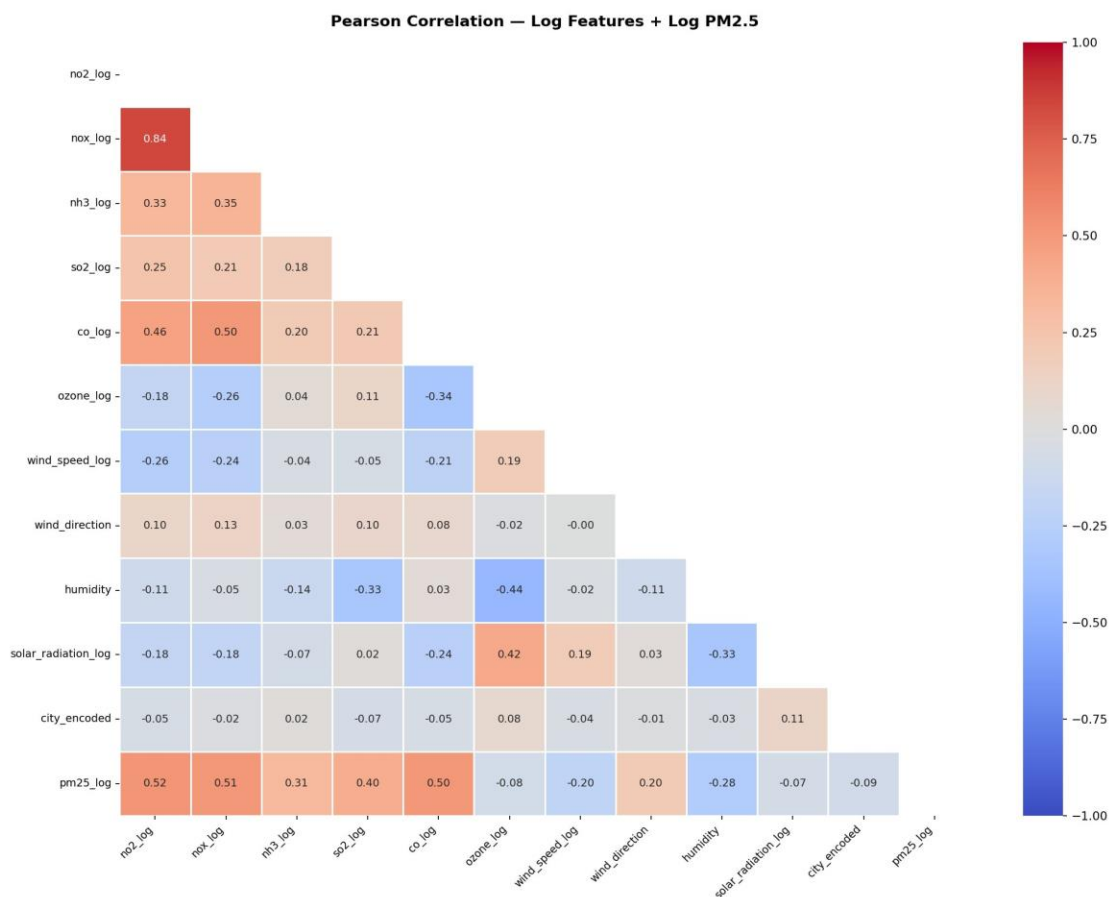


Figure 4: Correlation Heatmap — Log-transformed features vs log(PM2.5).

6.2 Variance Inflation Factor (VIF)

The Variance Inflation Factor measures multicollinearity among predictors. A VIF above 10 indicates problematic multicollinearity that can destabilise regression coefficient estimates. All features in both the raw and log-transformed feature sets showed VIF values below 5, confirming that multicollinearity is not a concern for these models.

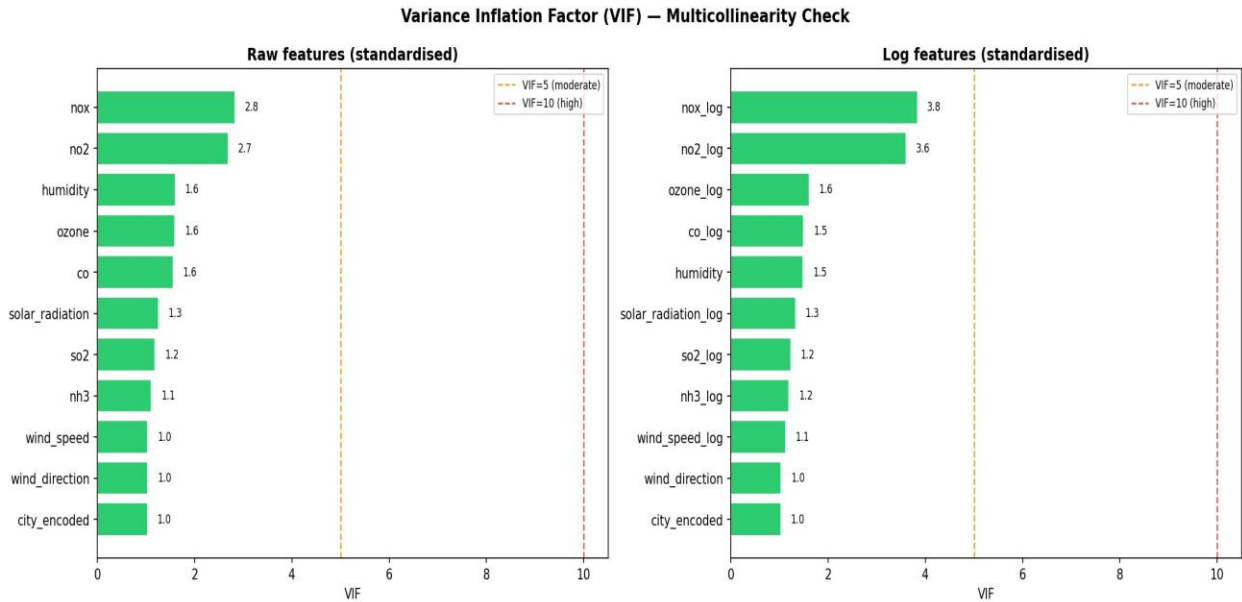


Figure 5: VIF Chart — All features remain below the VIF=5 threshold in both raw and log feature sets.

7. Model Building & Results

7.1 OLS Regression — Raw Features

Ordinary Least Squares regression was first applied to raw (untransformed) features to establish a linear baseline. Features were standardised using StandardScaler before fitting to ensure scale-invariant coefficient estimates. Statsmodels was used to obtain full regression summaries including p-values, confidence intervals, and information criteria.

The residual plot shows a pronounced funnel shape — variance of residuals increases with fitted values — which is a clear sign of heteroskedasticity. This violates the OLS homoskedasticity assumption and indicates that the raw-scale relationship between predictors and PM2.5 is not well captured by a linear model.

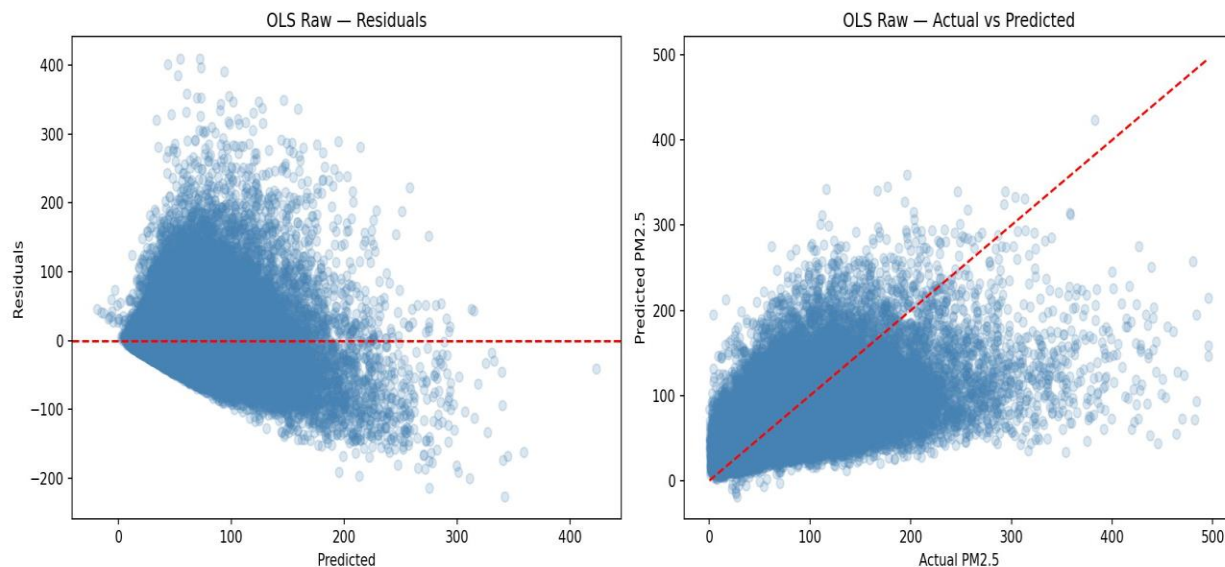


Figure 6: OLS Raw — Residual plot (left) and Actual vs Predicted (right).

7.2 OLS Regression — Log-Transformed Features

Applying $\log_{10}$ transforms to both features and target reduces heteroskedasticity by compressing the scale. The residuals are more symmetric around zero in log-space. However, back-transforming predictions to the original PM2.5 scale reveals that the model systematically underpredicts extreme pollution episodes. The lower R^2 (0.288) compared to OLS-Raw (0.427) occurs because the log target penalises large absolute errors less than the raw scale, but proportional errors are captured differently.

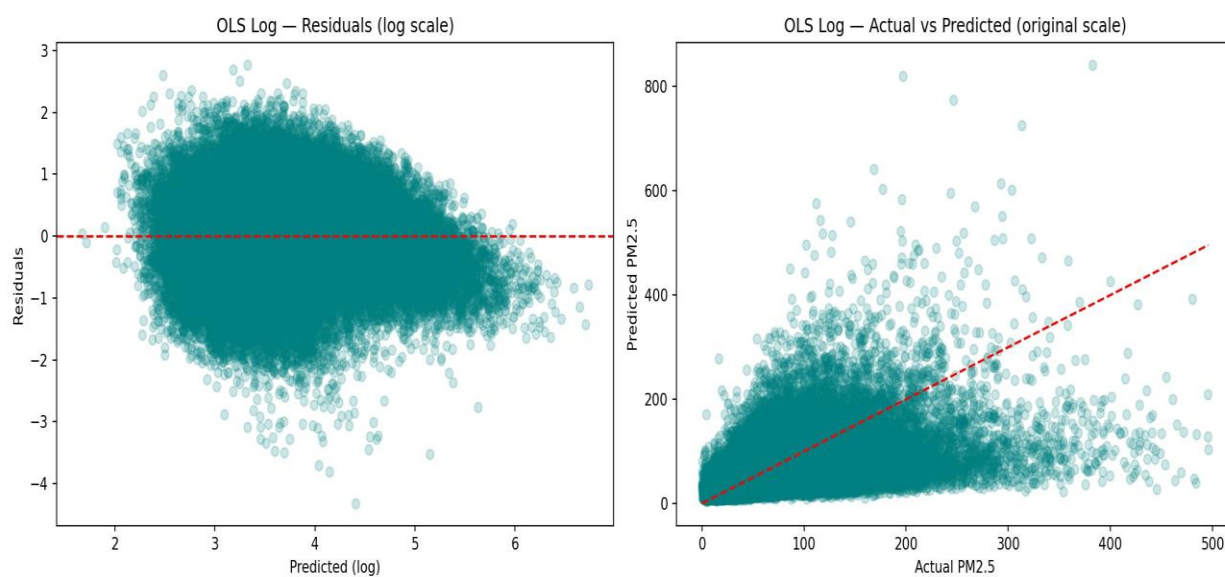


Figure 7: OLS Log — Residuals in log-space (left) and Actual vs Predicted on original scale (right).

7.3 Polynomial Regression (Degree 2)

To capture nonlinear relationships between pollutants and PM2.5, degree-2 polynomial features were generated from a subset of the six most correlated predictors: CO, NO₂, NO_x, SO₂, NH₃, and Humidity. With 6 input features, a degree-2 expansion generates 27 polynomial terms (original features, squared terms, and pairwise interactions).

StandardScaler was applied before polynomial expansion to prevent numerical instability from squaring large values. Ridge regression (L2 regularisation, $\alpha=1.0$) was used instead of plain OLS to handle the inherent multicollinearity introduced by polynomial terms and their interactions. Statsmodels was additionally used to compute AIC and BIC for model selection purposes.

Metric	Value
RMSE	37.52 $\mu\text{g}/\text{m}^3$
MAE	25.77 $\mu\text{g}/\text{m}^3$
R ²	0.470
Adjusted R ²	0.469
Polynomial Features	27 (degree 2)

Table 2: Polynomial Regression (Degree 2) performance metrics

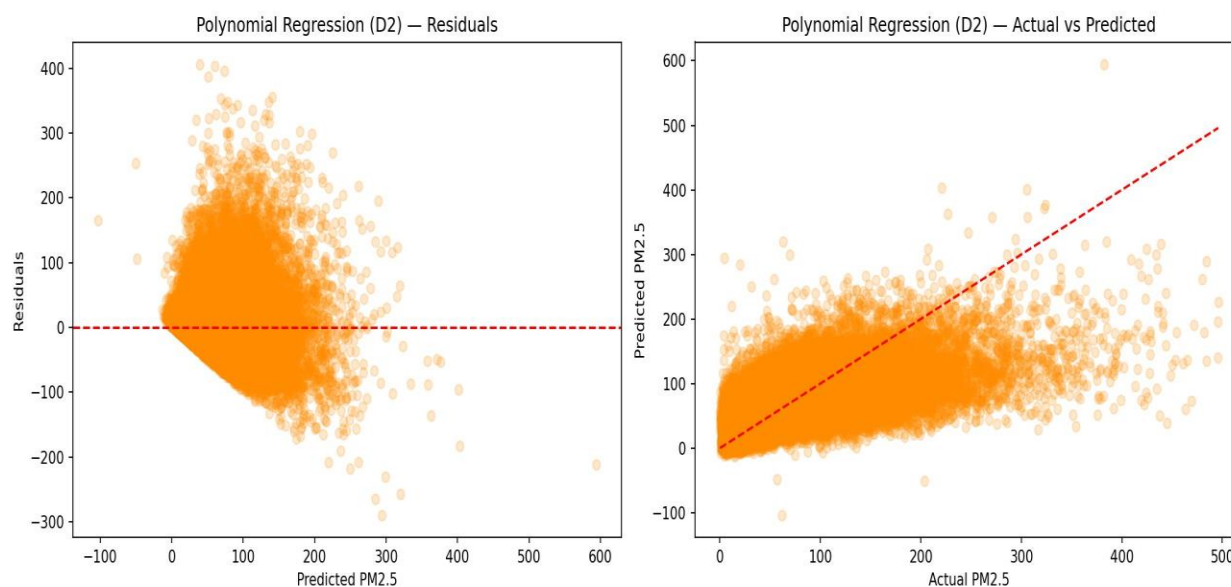


Figure 8: Polynomial Regression — Residuals (left) and Actual vs Predicted (right). Residual fan shape is reduced compared to OLS-Raw.

7.4 LightGBM

LightGBM is a gradient boosting framework that builds an ensemble of decision trees using histogram-based leaf-wise growth. It is particularly well-suited for tabular data with complex nonlinear interactions. The model was trained on log-transformed features with a log-transformed target ($\log_{1p}(\text{PM}_{2.5})$), with predictions back-transformed to the original scale for evaluation.

Key hyperparameters: learning rate 0.05, 63 leaves, feature fraction 0.8, bagging fraction 0.8, L1/L2 regularisation of 0.1 each, and early stopping at 50 rounds of no improvement. The model converged within 1,000 boosting rounds.

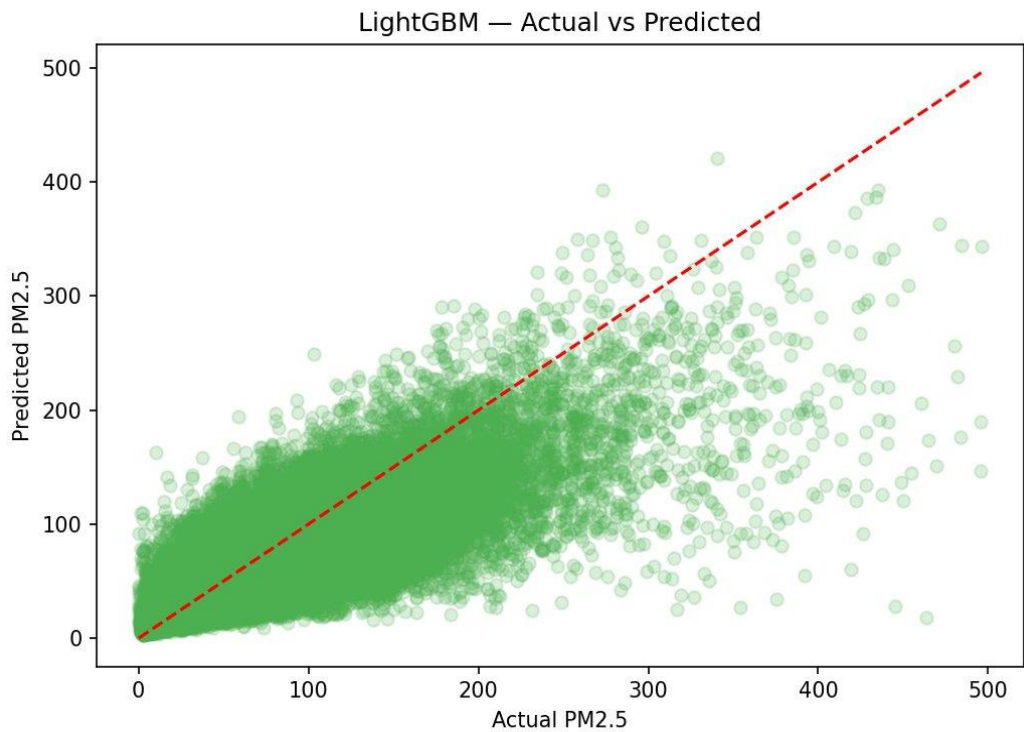


Figure 9: LightGBM — Actual vs Predicted (original PM2.5 scale). The model tracks the 45° reference line well across the full range.

Metric	Value
RMSE	28.40 $\mu\text{g}/\text{m}^3$
MAE	17.68 $\mu\text{g}/\text{m}^3$
R ²	0.696
Adjusted R ²	0.696

Table 3: LightGBM performance metrics (original PM2.5 scale)

7.5 XGBoost

XGBoost is another gradient boosting implementation, using a level-wise tree growth strategy with second-order gradient approximations. It was configured with the same depth, subsampling, and regularisation approach as LightGBM for a fair comparison. The same log-transformed feature set and target were used.

Key hyperparameters: 1,000 estimators with early stopping at 50 rounds, learning rate 0.05, max depth 6, subsample and colsample_bytree of 0.8, L1 regularisation 0.1, L2 regularisation 0.1.

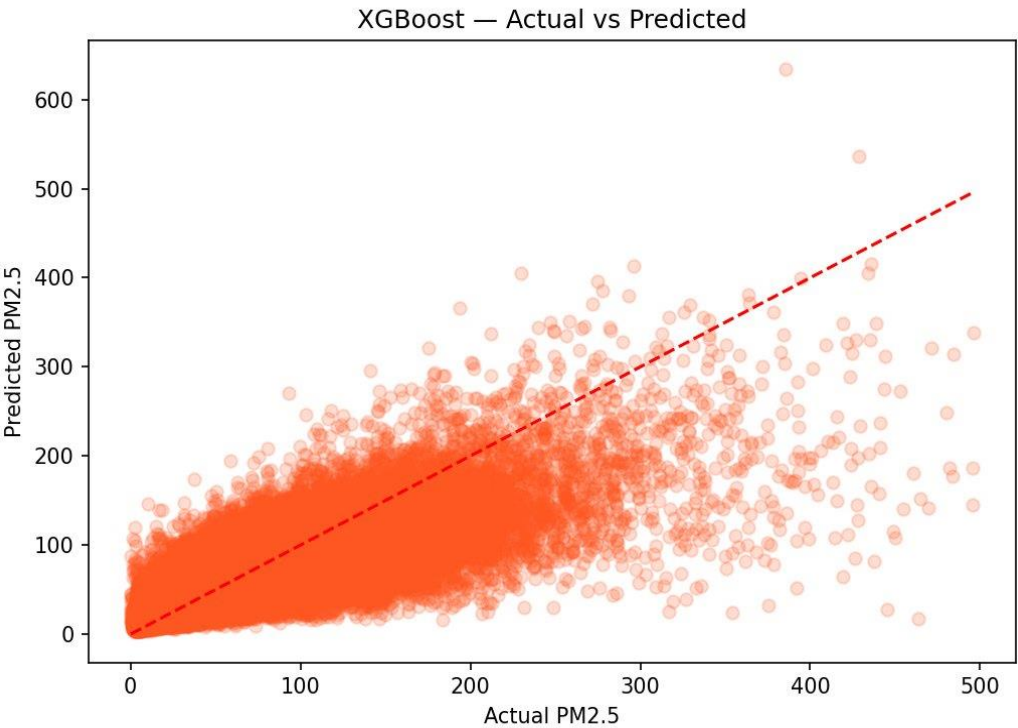


Figure 10: XGBoost — Actual vs Predicted (original PM2.5 scale).

Metric	Value
RMSE	29.53 $\mu\text{g}/\text{m}^3$
MAE	18.43 $\mu\text{g}/\text{m}^3$
R ²	0.672
Adjusted R ²	0.672

Table 4: XGBoost performance metrics (original PM2.5 scale)

8. SHAP Analysis — LightGBM Interpretability

SHAP (SHapley Additive exPlanations) values quantify how much each feature contributes to each individual prediction, grounded in cooperative game theory. A SHAP analysis was performed on a random sample of 25,000 test observations using the LightGBM model.

8.1 Feature Importance (Mean |SHAP|)

The mean absolute SHAP value gives a global ranking of feature importance — how much each feature moves predictions on average across all test points.

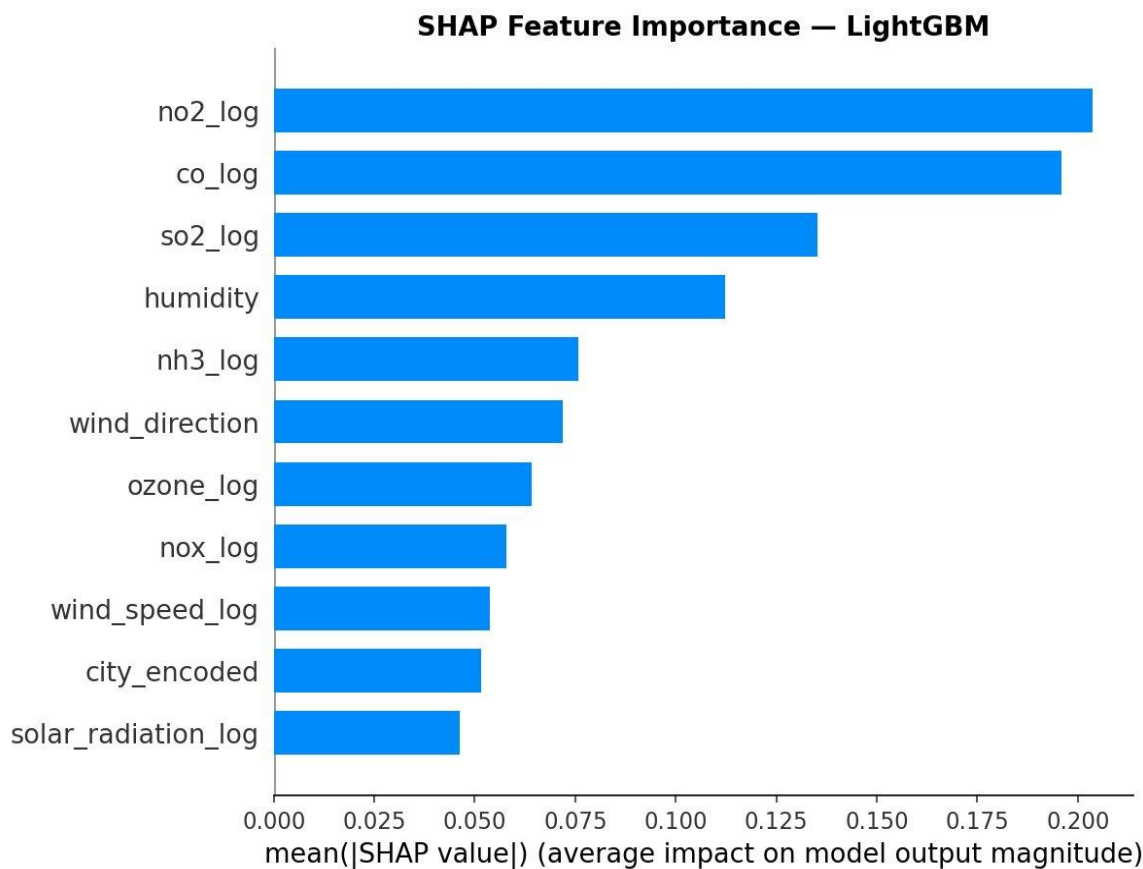


Figure 11: SHAP Feature Importance Bar Chart — Mean absolute SHAP value per feature.

NO₂ and CO are the two dominant predictors, each with a mean |SHAP| of approximately 0.20 on the log scale. SO₂ and Humidity are the next most important features, followed by NH₃ and Wind Direction.

8.2 SHAP Summary Dot Plot

The dot plot shows not only the magnitude but also the direction of each feature's impact. Red dots represent high feature values; blue dots represent low values. Points to the right of the centre line indicate a positive impact on the prediction (i.e., higher predicted PM2.5).

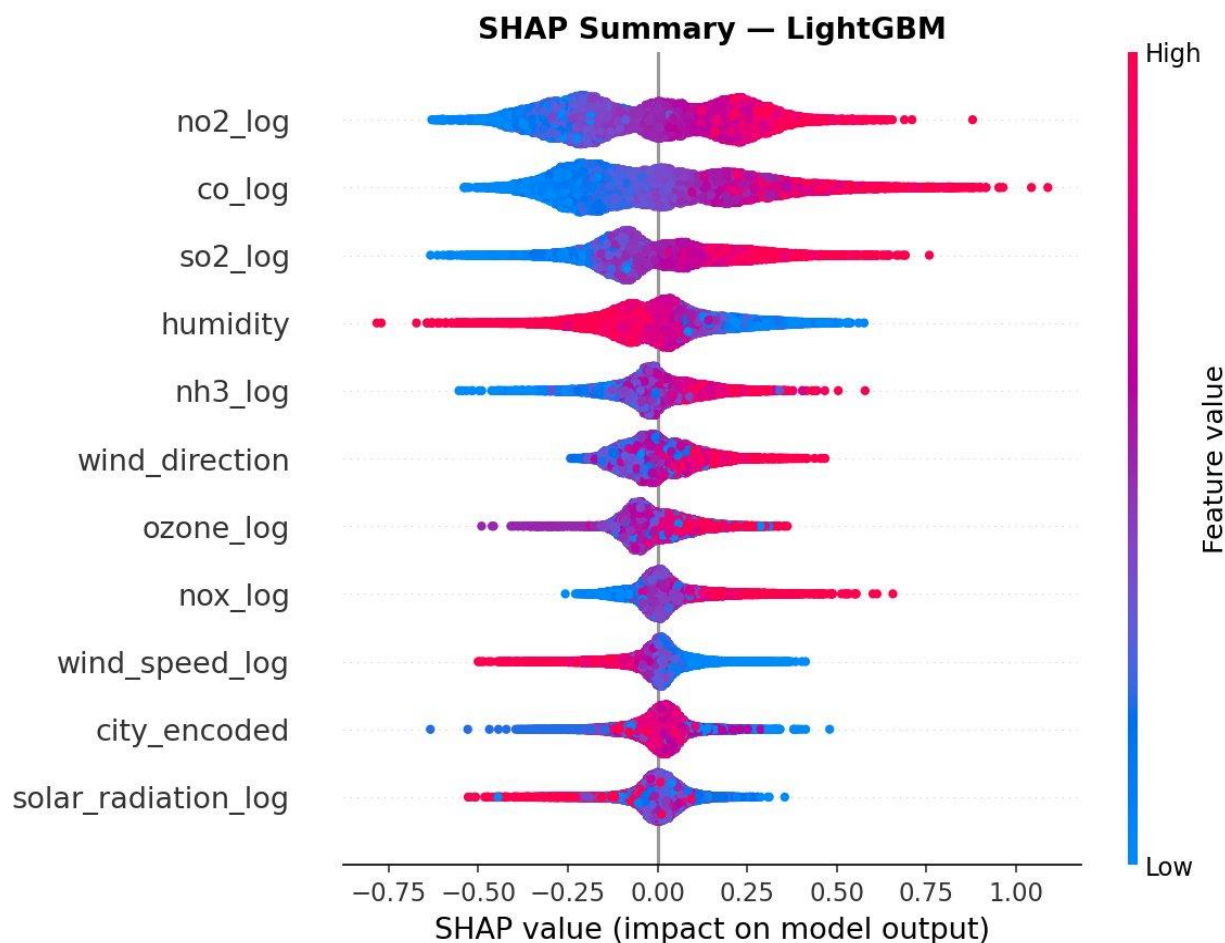


Figure 12: SHAP Summary Dot Plot — Distribution of SHAP values coloured by feature magnitude.

8.3 Key SHAP Findings

- NO₂ (no2_log): High NO₂ values strongly increase predicted PM2.5 — consistent with NO₂ being a major combustion co-product alongside fine particles.
- CO (co_log): Similarly, high CO increases PM2.5 — both are produced by incomplete combustion from vehicles and industry.
- SO₂ (so2_log): Positive driver — industrial emissions and coal burning release both SO₂ and fine particles together.
- Humidity: High humidity decreases PM2.5 — wet scavenging removes particulates from the atmosphere. Low humidity is associated with higher PM2.5.

- NH_3 (nh3_log): Moderate positive effect — ammonia can react with NO_x and SO_2 to form secondary aerosol particles.
- Wind Speed (wind_speed_log): High wind speed reduces PM2.5 through atmospheric dispersal.
- Ozone (ozone_log): Complex bidirectional effect, consistent with photochemical secondary aerosol formation.

9. Model Comparison

All models were evaluated on the original (untransformed) PM2.5 scale to allow direct comparison. For models trained on log-transformed targets, predictions were back-transformed using $\exp m1$ (i.e., $\exp(\text{prediction}) - 1$) before computing metrics.

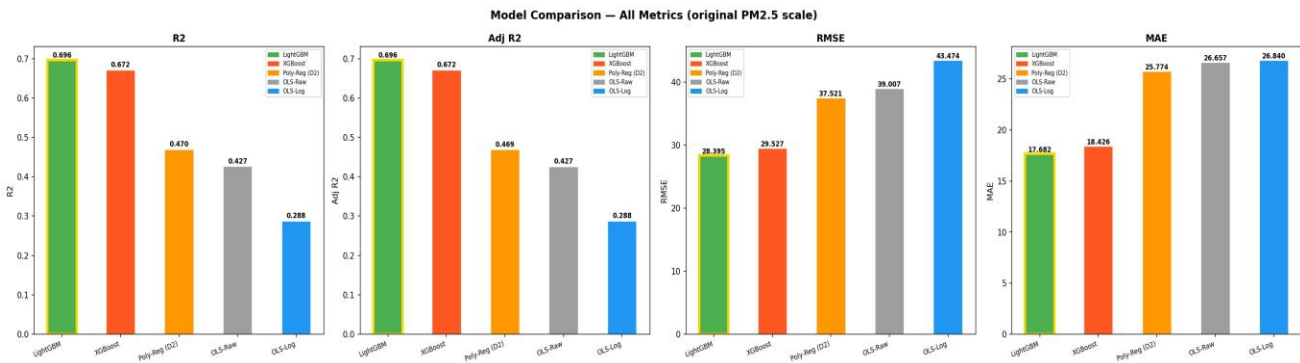


Figure 13: Bar chart comparison of all models across R^2 , Adjusted R^2 , RMSE, and MAE. Gold border marks the best model per metric.

Model	RMSE ($\mu\text{g}/\text{m}^3$)	MAE ($\mu\text{g}/\text{m}^3$)	R^2	Adj R^2
LightGBM	28.40	17.68	0.696	0.696
XGBoost	29.53	18.43	0.672	0.672
Poly-Reg (D2)	37.52	25.77	0.470	0.469
OLS-Raw	39.01	26.66	0.427	0.427
OLS-Log	43.47	26.84	0.288	0.288

Table 5: Final model comparison — all metrics on the original PM2.5 scale

9.1 Discussion

The results clearly demonstrate that tree-based ensemble models capture significantly more predictive signal than linear approaches. The gap between Polynomial Regression ($R^2=0.470$) and LightGBM ($R^2=0.696$) suggests that three-way and higher-order interactions, as well as threshold effects (e.g., PM2.5 spikes sharply above certain CO thresholds), are not captured even by quadratic feature expansion.

The OLS-Log model performs worse than OLS-Raw by R^2 (0.288 vs 0.427), though its MAE is marginally lower (25.77 vs 26.66). This is expected: the log transform compresses large errors, making the model appear better in relative terms but worse in absolute RMSE, which is dominated by the large PM2.5 events that matter most for air quality warnings.

Polynomial Regression bridges the gap between OLS and boosting models by capturing some nonlinearity, particularly the interaction between CO and NO₂ which SHAP analysis identifies as the two dominant predictors.

10. Conclusion & Future Work

10.1 Conclusions

- PM10 was correctly removed from the feature set due to its near-perfect correlation ($r=0.92$) with PM2.5, which would have constituted data leakage.
- No multicollinearity was detected among the retained features — all VIF values were below 5 in both raw and log feature spaces.
- Log transformation reduced skewness in pollutant distributions and improved correlation structure, though it did not convert the relationship to linear.
- Polynomial Regression ($R^2=0.470$) conclusively outperformed both OLS variants ($R^2=0.427$ and 0.288), confirming the nonlinear nature of PM2.5 generation.
- LightGBM achieved the best overall performance ($R^2=0.696$, RMSE=28.40 $\mu\text{g}/\text{m}^3$), followed by XGBoost ($R^2=0.672$, RMSE=29.53 $\mu\text{g}/\text{m}^3$).
- SHAP analysis identified NO₂ and CO as the primary drivers of PM2.5 elevation, with humidity and wind speed acting as dispersal mechanisms.

10.2 Future Work

- Temporal modelling: The current models treat each observation independently. Incorporating time-series structure — lagged features, rolling averages, or LSTM networks — could capture the autocorrelation in pollution levels.
- Spline / GAM models: Generalised Additive Models would allow each feature to have its own smooth nonlinear curve, providing better interpretability than polynomial features while outperforming linear OLS.
- Hyperparameter tuning: Bayesian optimisation (e.g., Optuna) on LightGBM/XGBoost could squeeze further performance gains.
- Spatial features: Incorporating station latitude/longitude and spatial proximity to industrial zones could capture geographic pollution sources.
- Secondary aerosol formation: Including features for atmospheric chemistry (e.g., humidity $\times$ NH₃ interaction) could model secondary PM2.5 formation from gas-phase precursors.

Appendix: Additional Visualisations

A.1 VIF Chart with PM10 (Before Removal)

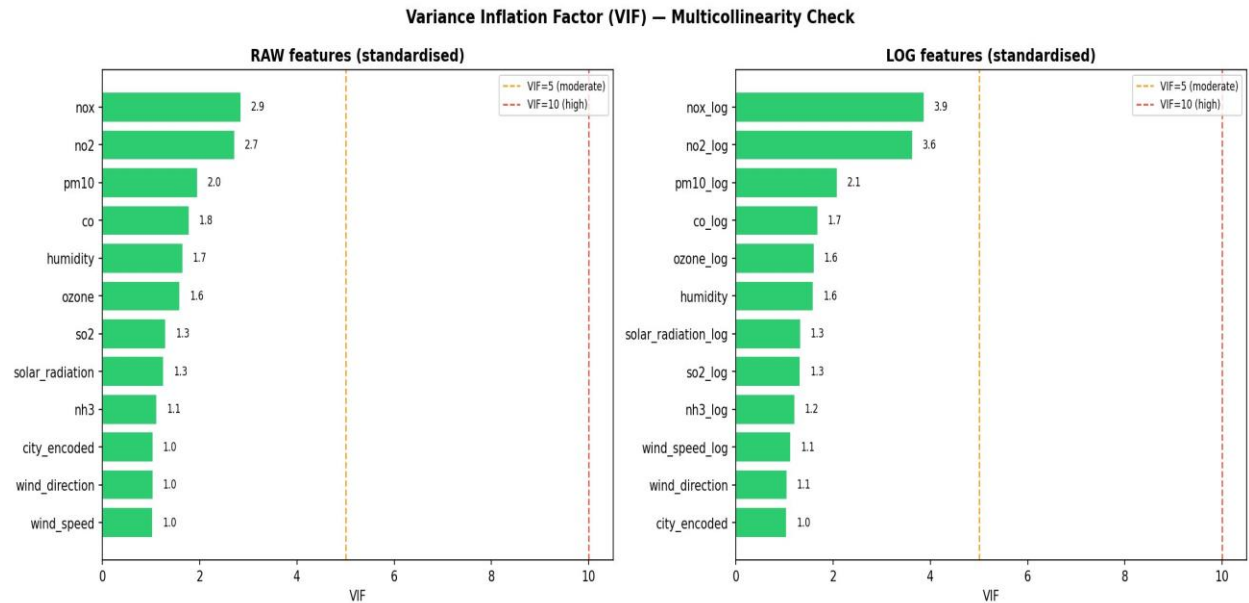


Figure A1: VIF chart when PM10 was included — PM10 had a VIF of 2.0 (raw) and 2.1 (log), not high enough to flag via VIF alone, but the correlation analysis ($r=0.92$ with PM2.5) identified it as a near-leakage feature.